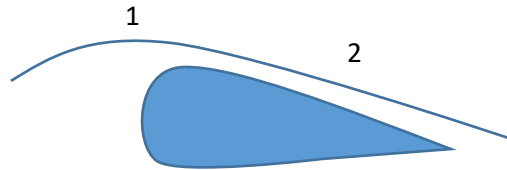


Quiz #15

Isentropic flow relationships are used for studying the airflow around an airfoil during high-speed flight. The flow relationships, along with the ideal gas law, let you compare one point on a streamline to another point – see the figure below.



Ideal gas law (for any point): $P = \rho RT$ ($R=1716 \text{ ft} - \text{lb}/\text{slug}^\circ\text{R}$)

Isentropic flow equation: $\frac{P_2}{P_1} = \left(\frac{T_2}{T_1}\right)^{3.5}$

Energy balance: $C_p T_1 + 0.5V_1^2 = C_p T_2 + 0.5V_2^2$ ($C_p = 6006 \text{ ft} - \text{lb}/\text{slug}^\circ\text{R}$) (Note: velocities must be in ft/s)

Assume that point 1 has the following conditions: pressure (P_1) = 2116 lb/ft², velocity (V_1) = 500 miles/hr, and density (ρ_1) = 0.002377 slug/ft³. Find the temperature (T_2) and velocity (V_2) at point 2 for pressure (P_2) values of 1000 to 2000 lb/ft² in increments of 10. Plot the following: P_2 vs V_2 , T_2 vs. P_2 , T_2 vs. V_2 . Create a modular program as follows:

1. Create and call (from the script) the following functions:
 - a. Inputs.m – This function initializes and returns the point 1 values and any other needed constants. Use input statements to collect values from the user.
 - b. Ideal_gasLaw.m – this function calculates the temperature at point 1.
 - c. Isentropic.m – this function uses the isentropic relationship to calculate the temperature at point 2.
 - d. Velocity.m – this function uses the energy balance equation to calculate the velocity at point 2.
 - e. Plot_vel_vs_press.m – this function creates the three plots. The plots should be in the format shown below.

Your final output should be a plot like the one shown (the plot shown has been rotated to fit it on the page). Note: The main script is used only to initialize variables and call the functions. Do not do calculations in the main script.



